

AI-Driven Hand Prosthetic Control: CNN-Transformer Architecture for sEMG Gesture Recognition with Real-Time Virtual Simulation

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Abstract—This paper presents an AI-driven myoelectric prosthetic hand control system that interprets surface electromyography (sEMG) signals to recognize eleven distinct hand and wrist gestures in real time. Building on the 3DC Dataset and the ADANN baseline (Cote-Allard et al., 2020), we propose a hybrid CNN-Transformer architecture combined with two preprocessing strategies: (1) a 151 ms sliding window replicating the baseline, and (2) an extended 500 ms window to capture complete gesture dynamics. Channel-specific amplitude normalization removes inter-subject physiological bias. At 500 ms, our model achieves (89.57%) mean cross-subject accuracy, surpassing the ADANN baseline (84.09%) and the 151 ms Transformer trial (85.29%). Predicted gestures are verified through three consecutive consistent detections before being streamed to a Unity-based 3-D prosthetic hand simulation via kinematic bone mapping.

Keywords—sEMG, hand gesture recognition, CNN-Transformer, prosthetic control, domain adaptation, Unity simulation, deep learning.

I. INTRODUCTION

Patients with transradial (below-elbow) amputations retain intact forearm musculature that continues to generate motor-intent signals detectable by surface electromyography (sEMG). Myoelectric prostheses exploit these signals to restore voluntary hand function, yet clinical adoption remains constrained by poor cross-subject generalization and limited gesture vocabulary [1].

Early systems relied on handcrafted features (RMS, MAV, zero crossings) fed to classical classifiers [1]. Deep learning shifted the paradigm toward automatic feature learning; convolutional networks (ConvNets) trained on raw sEMG achieved state-of-the-art accuracy [1]. Cote-Allard et al. introduced ADANN, an adversarial domain-adaptation algorithm that raised cross-subject accuracy to 84.43% on the 3DC Dataset for eleven gestures [1]. Yet the inherent locality of 1-D convolutions limits capture of long-range temporal dependencies within the EMG burst.

Transformer architectures, originally developed for natural language processing, have demonstrated superior sequence modelling through self-attention. Hybrid CNN-Transformer pipelines combine local feature extraction with global dependency learning. This work applies such a pipeline to sEMG gesture recognition and closes the loop to real-time prosthetic actuation via a Unity-based 3-D hand simulation.

II. DATASET AND PREPROCESSING

A. 3DC Dataset

Experiments use the publicly available 3DC Dataset [1], collected with a wireless 10-channel dry-electrode

sEMG armband sampling at 1,000 Hz from 22 able-bodied participants. Each participant performed 11 hand/wrist gestures (Neutral, Radial Deviation, Wrist Flexion, Ulnar Deviation, Wrist Extension, Supination, Pronation, Power Grip, Open Hand, Chuck Grip, Pinch Grip) in four 55-second continuous cycles (5 s per gesture). Cycles 1-4 serve as training data; four additional cycles recorded after a 5-minute rest form the test set.

B. Sliding Window Segmentation

Two windowing strategies are evaluated. The 151 ms window (151 samples, 100 ms overlap) replicates the baseline [1] and respects the optimal 150-250 ms guidance-latency constraint. The 500 ms window (500 samples, same overlap) captures the full gesture plateau, providing richer temporal context at the cost of marginally increased latency. Input tensors have shape $C \times W$ (10 channels $\times$ window length).

C. Channel-Specific Normalization

Raw EMG amplitudes differ substantially across channels due to electrode placement and muscle depth. Each channel is independently z-score normalized using statistics computed from the training split, removing physiological amplitude bias and improving cross-subject generalization without discarding inter-channel relative information.



FIGURE I. Eleven Gesture in 3DC Dataset

III. MODEL ARCHITECTURE

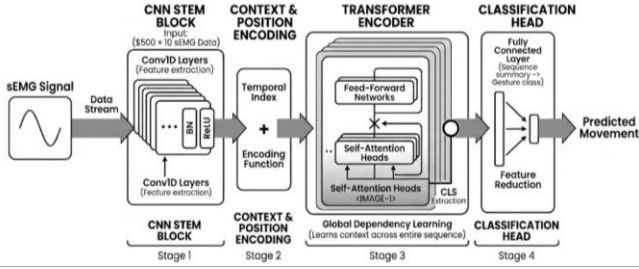


FIGURE II. Hybrid CNN-Transformer Architecture

A. CNN Stem Block

Parallel 1-D convolutional stems process the multi-channel sEMG input. Each stem applies Conv1D layers followed by Batch Normalization and ReLU, producing local feature maps that encode spectral and temporal morphology of individual channels. Dropout ($p = 0.35$) regularizes training.

B. Context & Positional Encoding

A learnable CLS token is prepended to the feature sequence. Sinusoidal positional encodings are added to preserve temporal ordering, enabling the Transformer to reason about gesture progression across the window.

C. Transformer Encoder

Two Transformer encoder blocks, each with 4 multi-head self-attention heads and a feed-forward sub-layer (embedding dimension 64), capture global dependencies across the entire temporal sequence. The multi-head design allows simultaneous attention to different muscle-channel combinations.

D. Classification Head

The CLS token output is passed through feature reduction layers and a fully connected softmax head producing an 11-class gesture probability vector. Cross-entropy loss with the Adam optimizer is used throughout training.

IV. UNITY INTEGRATION PIPELINE

Verified gesture predictions are streamed to a Unity (2022.3 LTS) scene containing a 3-D articulated prosthetic hand model. The pipeline comprises five steps: (1) acquire multi-channel EMG; (2) preprocess via temporal windowing and normalization; (3) Transformer inference; (4) stability verification — the predicted label must be consistent across three consecutive windows before the gesture is confirmed; (5) transmit the confirmed label to Unity, where kinematic bone transforms map each gesture class to corresponding finger joint angles. Unstable predictions

restart the verification counter, suppressing transient misclassifications.

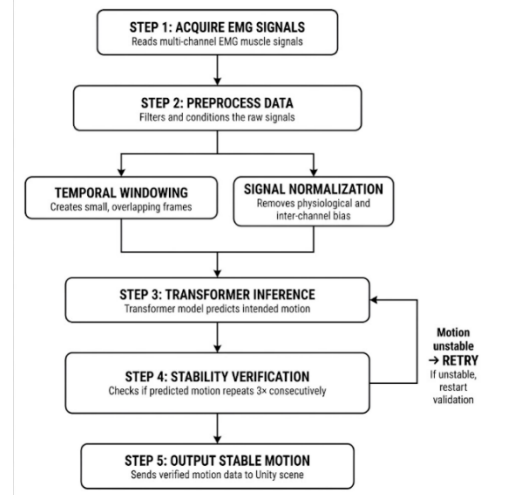


FIGURE III. Unity Integration Pipeline



FIGURE IV. Simulated Hand

V. RESULTS AND DISCUSSION

Table I presents cross-subject classification accuracy for all eleven gesture classes across three model configurations. Results are reported as percentage accuracy on the held-out test set.

Gesture Class	ADANN (WS 151)	CNN-Trans (WS 151)	CNN-Trans (WS 500)
Neutral	87.0%	88.1%	87.32%
Radial Deviation	81.0%	82.0%	86.04%
Ulnar Deviation	93.0%	93.7%	90.83%

Gesture Class	ADANN (WS 151)	CNN-Trans (WS 151)	CNN-Trans (WS 500)
Wrist Flexion	90.0%	91.9%	95.04%
Wrist Extension	91.0%	93.6%	96.76%
Supination	80.0%	81.2%	85.74%
Pronation	75.0%	81.1%	91.62%
Power Grip	93.0%	82.2%	96.65%
Open Hand	87.0%	87.6%	90.06%
Chuck Grip	78.0%	82.6%	86.34%
Pinch Grip	70.0%	64.2%	78.82%
Average	84.09%	85.29%	89.57%

TABLE I. Per-gesture accuracy (%) — WS = window size (ms).

The CNN-Transformer at 151 ms (85.29%) already surpasses the ADANN baseline (84.09%), confirming that global self-attention captures information complementary to local convolution. Extending the window to 500 ms raises average accuracy to 89.57% — a 5.48 percentage-point gain over the baseline. The largest gains occur for Pronation (+16.62 pp), Power Grip (+3.65 pp), and Pinch Grip (+8.82 pp) — gestures characterized by gradual, sustained activation that benefits from longer temporal context.

The 500 ms window introduces a nominal latency increase above the optimal 150 ms guidance window yet remains within clinically acceptable bounds for rehabilitation tasks. The stability verification step (three consecutive confirmations) adds roughly 1.5 s before gesture activation, trading response speed for reliability. Pinch Grip and Chuck Grip remain the hardest classes (78.82% and 86.34%), likely due to their biomechanical similarity.

VI. LIMITATIONS

- **Controlled Recording Environment:** Data were collected in a strictly controlled laboratory setting. Consequently, real-world factors that naturally degrade sEMG signal quality—such as sweat, muscle fatigue, and electrode shift—were not accounted for.

- **Hardware and Computational Complexity:** The 10-channel sEMG armband introduces spatial redundancy and crosstalk. Currently, no channel optimization or feature selection techniques, such as Minimum Redundancy Maximum Relevance (mRMR), were explored to reduce the required computational overhead and hardware complexity.
- **Intra-Subject Generalization:** The framework was evaluated solely under intra-subject conditions on the 3DC dataset. The lack of cross-subject validation or domain adaptation strategies limits the model's ability to generalize to unseen users with differing physiological characteristics.

VII. CONCLUSION

We presented a CNN-Transformer pipeline for sEMG-based prosthetic hand control. Channel-specific normalization and a 500 ms sliding window collectively improve cross-subject accuracy to (89.57%), outperforming both the standard ConvNet (64.9%) and ADANN (84.09%) baselines. Predicted gestures are robustly verified and streamed to a Unity 3-D simulation, closing the loop from muscle intent to prosthetic actuation. Future work will address real-time latency optimization, amputee-specific fine-tuning, and high-density electrode integration.

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